Arista Access Node KR-ACC-24 Routing Protocol Flap Resolution SOP

Node: KR-ACC-24 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_9289

Description

This Standard Operating Procedure (SOP) addresses the failure scenario of a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on the Arista Access Node KR-ACC-24 at KR Puram Access Site. The underlying issue lies in the complex interplay between the routing protocols and the hardware mechanisms. The BGP route churn occurs due to the dynamic nature of BGP routing protocol, which continuously updates the routing tables based on the best path to reach a destination network. The OSPF adjacency loss is caused by the OSPF protocol's reliance on hello packets to establish and maintain adjacencies between routers. ISIS LSP flooding occurs due to the high frequency of LSP updates, which can overwhelm the router's processing capabilities. The Arista Access Node's hardware, equipped with high-speed interfaces and advanced routing capabilities, exacerbates the issue by accelerating the rate at which routing protocols converge and reconverge.

Pre-requisite Checks

- Verify that the KR Puram Access Site is accessible via remote hands to perform necessary checks and interventions.

- Check the status of the redundant link to ensure it is operational and can take over in case of a failure.

- Confirm that the Arista Access Node KR-ACC-24 is powered on and operational.

- Review the system logs to identify any recent configuration changes or issues that may be contributing to the failure.

Potential Root Causes

Cause	Probability
BGP route churn caused by frequent route updates	High
OSPF adjacency loss due to hello packet loss or misconfiguration	Medium
ISIS LSP flooding due to high frequency of LSP updates	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP router identifier 10.10.10.1, local AS number 100, Description: This command displays a summary of the BGP routing tables, including the number of active routes and the number of neighbors. The output helps identify if there are any issues with BGP route updates or neighbor relationships.

Command: show ip ospf neighbor, Expected Output: Neighbor 10.10.10.2, interface Ethernet1/1, state FULL, Description: This command displays
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the OSPF neighbor relationships, including the interface used to establish the adjacency and the state of the adjacency. The output helps identify if there are any issues with OSPF adjacency establishment or maintenance.

Command: show isis database, Expected Output: ISIS LSP database, level 1, 2 entries, Description: This command displays the ISIS LSP database, including the number of entries and the level of the LSPs. The output helps identify if there are any issues with ISIS LSP flooding or LSP updates.
```

Resolution Steps

Step 1: Step: Clear BGP route updates, Command: clear ip bgp *, Description: This command clears all BGP route updates and forces the router to re-establish the routing table. This step helps resolve BGP route churn issues caused by frequent route updates.

Step 2: Step: Re-establish OSPF adjacency, Command: clear ip ospf neighbor *, Description: This command clears the OSPF neighbor relationships and forces the router to re-establish the adjacency. This step helps resolve OSPF adjacency loss issues caused by hello packet loss or misconfiguration.

Step 3: Step: Reduce ISIS LSP updates, Command: isis database-filter out level1, Description: This command filters out level 1 LSPs from the ISIS LSP database, reducing the frequency of LSP updates and helping to resolve ISIS LSP flooding issues.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP router identifier 10.10.10.1, local AS number 100, Description: This command verifies that the BGP routing tables are stable and that the number of active routes and neighbors are within expected ranges.

Command: show ip ospf neighbor, Expected Output: Neighbor 10.10.10.2, interface Ethernet1/1, state FULL, Description: This command verifies that the OSPF neighbor relationships are stable and that the adjacency state is FULL.

Command: show isis database, Expected Output: ISIS LSP database, level 1, 1 entry, Description: This command verifies that the ISIS LSP database is stable and that the number of entries is within expected ranges.
```

Roll-back Procedures

- Step: Restore original BGP configuration, Command: copy running-config startup-config, Description: This command restores the original BGP configuration from the startup configuration file, rolling back any changes made during the resolution process.

- Step: Restore original OSPF configuration, Command: copy running-config startup-config, Description: This command restores the original OSPF configuration from the startup configuration file, rolling back any changes made during the resolution process.

- Step: Restore original ISIS configuration, Command: copy running-config startup-config, Description: This command restores the original ISIS configuration from the startup configuration file, rolling back any changes made during the resolution process.